Sindh Science Learning Portal

Class 5 · Science

Chapter 2

Microorganisms



Learn what microorganisms are, how bacteria, viruses and fungi differ, and how they can be useful or harmful.

Following the Sindh Textbook Board curriculum

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CHAPTER 2

Microorganisms

What we will learn Learn what microorganisms are, how bacteria, viruses and fungi differ, and how they can be useful or harmful.

1 What are microorganisms?

Some living things are so small that we cannot see them with our eyes. They are called **microorganisms** or **microbes**.

They can only be seen with a **microscope**. A microscope makes tiny things look much bigger.

Microorganisms are found everywhere: in soil, air and water, on plants and animals, and inside and outside our bodies.

There are three main groups: **bacteria**, **viruses** and **fungi**.

2 Bacteria, viruses and fungi

- **Bacteria**: living things made of one cell. Some are round, some are rod-shaped and some are spiral. They feed, breathe and grow by dividing.
- **Viruses**: the smallest microbes. They can only be seen with a very powerful microscope. A virus can only grow inside the cells of a living thing. Colds, flu, polio and measles are caused by viruses.
- **Fungi**: some fungi are tiny and have one cell, like **yeast**. Some are large, like **mushrooms**. The green or black **mould** on old bread is also a fungus.

Fungi cannot make their own food. They take food from dead or rotting things.



Picture: Jan Myšák · CC BY-SA 4.0 · Wikimedia Commons

Diagram: the parts and what they do



Microorganisms (خوردبيني جاندار · خوردبینی جاندار)
Tiny living things that can only be seen with a microscope.

Bacteria (بيكتيريا · بیکتیریا)
One-celled. Some make yoghurt; some cause cholera and typhoid.

Viruses (وائرس · وائرس)
The smallest. They grow only inside living cells. They cause colds, flu and polio.

Fungi (فنگس · فنجائی)
Yeast, mould and mushrooms. Yeast makes dough rise; penicillin comes from a fungus.

3 Useful microorganisms

Most microorganisms are not harmful. Many are our friends:

- **Decomposing**: bacteria and fungi break down dead plants, dead animals and rubbish and mix them into the **soil**. This makes the soil rich.
- **Making food**: **yeast** makes dough rise, so naan and bread become soft. Bacteria turn milk into **yoghurt** and cheese.
- **Digestion**: some bacteria living in our intestines help us digest food.
- **Medicines**: some **antibiotics** are made from microbes. **Penicillin** is made from a fungus. **Vaccines** (like polio drops) protect us from diseases.

4 Harmful microorganisms and staying safe

Some microbes **spoil** food and spread diseases. A disease caused by microbes is called an **infection**.

- Caused by bacteria: **cholera**, **typhoid** and some kinds of diarrhoea.
- Caused by viruses: colds, flu, measles and **dengue** (the dengue virus is spread by mosquitoes).
- Caused by fungi: athlete's foot (itching between the toes) and some skin diseases.

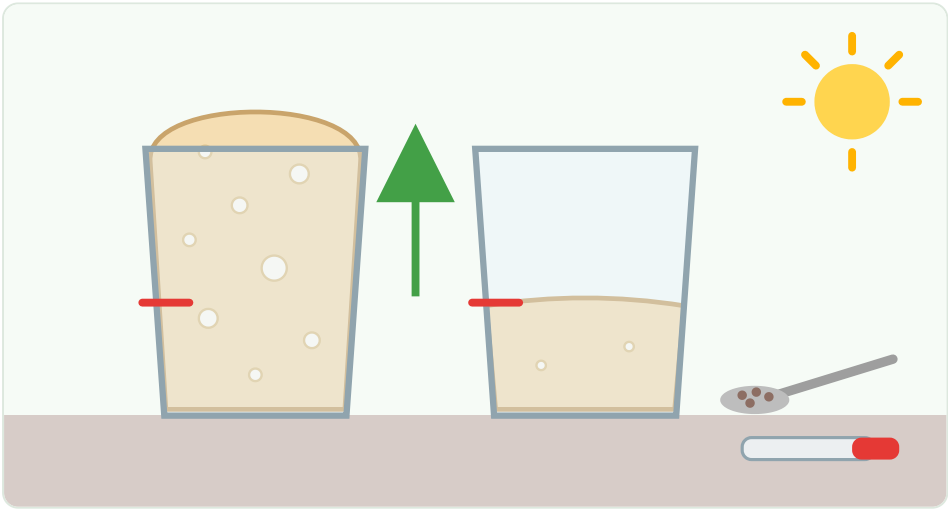
Microbes enter the body through the mouth (dirty food and water), the nose (breathing) and cuts in the skin.

Stay safe: **wash your hands** with soap, boil drinking water, keep food covered, keep your surroundings clean, remove standing water and get **vaccinated**.

Key words

English	سنڌي	اردو
microorganism	خوردبيني جاندار	خوردبینی جاندار
microscope	خوردبين	خوردبین
bacteria	بيڪٽيريا	بیکٹیریا
virus	وائرس	وائرس
fungi	فنگس	فنجائی
yeast	خمير	خمیر
mould	ڦوڪڙ	پھپھوندی
antibiotic	اينٽي بايوٽڪ	اینٹی بایوٹک
vaccine	ويڪسين	ویکسین
infection	انفيڪشن	انفیکشن
hygiene	صفائي	حفظان صحت

Experiment: Does yeast make dough rise?



You need

- Two small bowls of flour
- Half a spoon of yeast
- One spoon of sugar
- Lukewarm water
- Two glasses and a marker

Steps

1. Mix the yeast and sugar into the first bowl of flour. Add nothing to the second bowl.
2. Add a little lukewarm water to both and knead two soft doughs.
3. Press each dough into its own glass and mark its height with the marker.

4. Cover both glasses and keep them in a warm place for 2 hours.
5. Compare the height of each dough with its mark.

Safety: Use lukewarm water, not hot water. Ask an adult to help, and wash your hands afterwards.

What did we learn? The dough with yeast rose much more. Yeast is a tiny fungus that feeds on sugar and gives out a gas. The gas makes small bubbles in the dough, so it rises. This is an example of useful microbes.

Practice questions

Circle the correct answer

1. **What do we need to see microorganisms?**
   - **(a)** A microscope
   - **(b)** A telescope
   - **(c)** A mirror
   - **(d)** Only our eyes
2. **Where are microorganisms found?**
   - **(a)** Only in dirty water
   - **(b)** Everywhere: in soil, air, water and our bodies
   - **(c)** Only in hospitals
   - **(d)** Only on plants
3. **Which are the smallest microbes?**
   - **(a)** Mushrooms
   - **(b)** Bacteria
   - **(c)** Viruses
   - **(d)** Yeast
4. **What is the green mould on old bread?**
   - **(a)** A fungus
   - **(b)** A virus
   - **(c)** A plant
   - **(d)** Soil
5. **Which microorganism makes dough rise?**
   - **(a)** Virus
   - **(b)** Yeast
   - **(c)** Mosquito
   - **(d)** Cholera bacteria
6. **Penicillin is made from what?**
   - **(a)** A fungus
   - **(b)** A rock
   - **(c)** Sea water
   - **(d)** A virus
7. **What do we call breaking down dead leaves into the soil?**
   - **(a)** Infection
   - **(b)** Decomposition
   - **(c)** Vaccination
   - **(d)** Rising

8. **What spreads the dengue virus?**

(a) Honeybee (b) Mosquito

(c) Dog (d) Cat

9. **What is the best way to stay safe from germs?**

(a) Washing hands with soap before eating (b) Leaving food uncovered

(c) Drinking unboiled pond water (d) Keeping rubbish in the house

Write true or false

10. **All microorganisms are harmful.** _____

11. **A virus can only grow inside the cells of a living thing.** _____

12. **Bacteria turn milk into yoghurt.** _____

13. **Cholera is caused by a virus.** _____

14. **Boiling water kills the harmful germs in it.** _____

Match the columns

15. **Match each microbe with its group.**

1. Yeast		(a) Virus
2. Polio germ		(b) Bacteria
3. Yoghurt germs		(c) Fungus

Match the columns

16. **Match each disease with its cause.**

1. Typhoid		(a) Virus
2. Measles		(b) Fungus
3. Athlete's foot		(c) Bacteria

Worksheet: Microorganisms

Name: _____ Date: _____

A. Fill in the blanks

Word box: Viruses · Yeast · fungus · infection · microscope

1. An instrument that makes tiny things look bigger is called a _____ .
2. _____ are the smallest microbes.
3. _____ makes dough rise.
4. Penicillin is made from a _____ .
5. A disease caused by microbes is called an _____ .

B. Put ✓ for true and ✗ for false

1. ☐ Bacteria are made of one cell.
2. ☐ Mushrooms are a kind of virus.
3. ☐ Vaccines protect us from diseases.
4. ☐ Food left uncovered never gets germs.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Yeast	( ) Common cold
2. Bacteria	( ) Polio drops
3. Virus	( ) Naan and bread
4. Vaccine	( ) Yoghurt

D. Answer in short

1. Name the three groups of microorganisms.

_____

_____

2. Give two examples of useful microbes.

_____

_____

3. Write three ways to protect ourselves from germs.

_____

_____

E. Draw and label

Draw three boxes: bacteria, virus and fungus. In each box draw a picture and write one example.

FOR TEACHERS

Answer key

Chapter 2: Microorganisms

Practice questions

1. (a) A microscope
2. (b) Everywhere: in soil, air, water and our bodies
3. (c) Viruses
4. (a) A fungus
5. (b) Yeast
6. (a) A fungus
7. (b) Decomposition
8. (b) Mosquito
9. (a) Washing hands with soap before eating
10. False
11. True
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (c), 2 → (a), 3 → (b)

Worksheet

A. Fill in the blanks: 1. microscope · 2. Viruses · 3. Yeast · 4. fungus · 5. infection

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Common cold · (4) Polio drops · (1) Naan and bread · (2) Yoghurt

D. Answer in short:

1. Bacteria, viruses and fungi.
2. Yeast makes dough rise; bacteria turn milk into yoghurt.
3. Wash hands with soap, boil drinking water and keep food covered.

Picture credits

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